

CS 7545 F26, Homework 1

100 point assignment.

120 points possible.

Points above 100 are divided by four and become bonus points.

Maximum bonus 5 points.

Due September 17, 2026

Exercise 1 (20 points)

Suppose we independently draw n vectors $\{\vec{x}^{(i)}\}_{i=1}^n$ where each $\vec{x}^{(i)} \in \{-1, 1\}^d$ has entries independently sampled according to the following distribution:

$$\vec{x}_j^{(i)} \sim \begin{cases} -1 & \text{w.p. } 0.5 \\ 1 & \text{w.p. } 0.5 \end{cases} \quad (1)$$

Proposition 1. For any $n \geq 1$, $d \geq 1$, $0 < \delta < 1$, if $n \leq \sqrt{2\delta} \exp(d/16)$, then

$$\Pr\left(\forall i \neq j : \|\vec{x}^{(i)} - \vec{x}^{(j)}\| \geq \sqrt{d}\right) \geq 1 - \delta \quad (2)$$

Where $\|\cdot\|$ is the Euclidean norm (2-norm). Intuitively, even for n exponentially large in d , all vectors are separated with high probability.

Prove the proposition.

Exercise 2 (40 points)

Suppose $\{Z_i\}$ are n independent random variables on $\mathbb{R}$ with pdf

$$f_{Z_i}(z) = \frac{\exp(-|z|)}{2} \quad (3)$$

Define

$$X_i = \begin{cases} 0 & \text{if } |Z_i| > M \\ Z_i & \text{if } |Z_i| \leq M \end{cases} \quad (4)$$

Thus, X_i are bounded in $[-M, M]$ and $\Pr(X_i = 0) = \exp(-M)$.

A key quantity is the empirical average: $A = \frac{1}{n} \sum_{i=1}^n X_i$.

1. Prove that $\mathbb{E}[X_i] = 0$, and $\Pr(|A| \geq \epsilon) = 2\Pr(A \geq \epsilon)$ for any $\epsilon > 0$.
2. Apply Hoeffding's inequality to bound $\Pr(A \geq \epsilon)$ as a function of ϵ , M , and n .
3. Show that the variance is

$$\mathbb{E}[X_i^2] = 2 - (M^2 + 2M + 2)\exp(-M) \quad (5)$$

You don't need to show it, but from this, we can see that $\mathbb{E}[X_i^2] \leq 2$, and for $M \geq 10$, $\mathbb{E}[X_i^2] \geq 1.99$.

4. Apply Bernstein's inequality (using $\mathbb{E}[X_i^2] \leq 2$) to bound $\Pr(A \geq \epsilon)$ as a function of ϵ , M , and n .
5. Show that for $M \geq 2$, Bernstein's inequality is always tighter than Hoeffding's inequality. You can assume $\epsilon \leq M$, since otherwise, there is a trivial bound $\Pr(A \geq \epsilon) = 0$.
6. Show that for $0 < t < 1$,

$$\mathbb{E}[\exp(tX_i)] \leq \frac{1}{1-t^2} \quad (6)$$

Hint: you can use the identity that for $0 < t < 1$ and $0 \leq x \leq 1$, $x \leq \frac{1}{2} \left[\frac{x^{(1-t)}}{(1-t)} + \frac{x^{(1+t)}}{(1+t)} \right]$

7. Show that

$$\Pr(A \geq \epsilon) \leq \exp(-nf(\epsilon)) \quad (7)$$

where

$$f(\epsilon) = \ln \left(\frac{2}{\epsilon^2} \left(\sqrt{1 + \epsilon^2} - 1 \right) \right) + \left(\sqrt{1 + \epsilon^2} - 1 \right) \quad (8)$$

8. Show the following,

$$f(\epsilon) \geq \frac{\epsilon^2}{4 + \frac{1}{2}\epsilon^2} \quad (9)$$

and use it to conclude the bound from part 7 is always tighter than Bernstein's inequality from part 4, for $\epsilon \leq M$ (the non-trivial case).

Hint: set $s = \sqrt{1 + \epsilon^2} - 1$ and note $s(s + 2) = \epsilon^2$. Use a comparison of the derivatives of the two sides.

Exercise 3 (20 points)

In this exercise, you will prove a version of Azuma's inequality.

Let $S_0 = 0$, and $S_1, \dots, S_n$ be random variables where $\mathbb{E}[S_i - S_{i-1} | S_1, \dots, S_{i-1}] = 0$ for all $i \in [n]$ and values of $S_1, \dots, S_{i-1}$, and $|S_i - S_{i-1}| \leq c$ for all $i \in [n]$.

1. Prove the following for all $i \in [n]$ and $t > 0$,

$$\mathbb{E}[\exp(t(S_i - S_{i-1})) | S_1, \dots, S_{i-1}] \leq \exp\left(\frac{t^2 c^2}{2}\right) \quad (10)$$

Hint: fix values for $S_1, \dots, S_{i-1}$, and give a name to $S_i - S_{i-1} | S_1, \dots, S_{i-1}$. Then, follow the proof of Hoeffding's inequality from the course notes.

2. Prove the following for all $i \in [n]$ and $t > 0$,

$$\mathbb{E}[\exp(tS_i)] \leq \exp\left(\frac{t^2 c^2}{2}\right) \mathbb{E}[\exp(tS_{i-1})] \quad (11)$$

Hint: use the law of total expectation with $S_1, \dots, S_{i-1}$.

3. Prove the following for all $\epsilon > 0$,

$$\Pr(S_n \geq \epsilon n) \leq \exp\left(-\frac{n\epsilon^2}{2c^2}\right) \quad (12)$$

Exercise 4 (20 points)

Let $\mathcal{X}$ be an input space, D a data distribution over $\mathcal{X} \times \{-1, 1\}$, and H a finite set of hypotheses $h : \mathcal{X} \rightarrow \{-1, 1\}$. Define $L(h) = \Pr_{(x,y) \sim D}(h(x) \neq y)$. Assume

$$\forall h \in H : L(h) \leq p \quad (13)$$

for some $p \leq 1/2$. Let S be an i.i.d. sample of size n from the data distribution and $\hat{L}_S(h) = \frac{1}{|S|} \sum_{(x,y) \in S} \mathbf{1}[h(x) \neq y]$.

1. Apply Bernstein's inequality (using the max loss from equation 13) and union bound to bound

$$\Pr(\exists h \in H : L(h) - \hat{L}_S(h) \geq \epsilon) \quad (14)$$

Hint: what is the variance of a Bernoulli random variable?

2. Use the result from part 1 to show that with probability at least $1 - \delta$, for any $h \in H$,

$$L(h) \leq \hat{L}_S(h) + \sqrt{\frac{2p(1-p)(\ln(|H|) + \ln(1/\delta))}{n}} + \frac{2}{3n}(\ln(|H|) + \ln(1/\delta)) \quad (15)$$

Exercise 5 (20 points)

Let L and $\hat{L}_S$ be the population and empirical zero-one (i.e. misclassification) loss, as in exercise 4. Let H be a finite set of hypotheses. Let $h^* \in \arg \min_{h \in H} L(h)$ and $\hat{h}_S \in \arg \min_{h \in H} \hat{L}_S(h)$. Here, we will explore connections between expected error and high probability bounds on the error.

Define $E = \mathbb{E}_S[L(\hat{h}_S) - L(h^*)]$.

1. First, we use E to get a probability bound. Prove that for any $E \neq 0$ and $\delta > 0$, $\Pr(L(\hat{h}_S) - L(h^*) > E/\delta) \leq \delta$.
2. Now, suppose we knew that for some ϵ and δ , $\Pr(L(\hat{h}_S) - L(h^*) \geq \epsilon) \leq \delta$. Show that this implies $E \leq \epsilon + \delta$. Hint: note that for a non-negative random variable X , $\mathbb{E}[X] = \int_0^\infty \Pr(X \geq x) dx$

3. We can do better by using many ϵ, δ pairs. From Hoeffding's Inequality, union bound, and uniform convergence, we know that $\Pr(L(\hat{h}_S) - L(h^*) \geq \epsilon) \leq 2|H| \exp(-n\epsilon^2/2)$. Prove that:

$$E \leq \sqrt{\frac{2 \ln(2|H|)}{n}} \left(1 + \frac{1}{2 \ln(2|H|)} \right) \quad (16)$$

Hint: you can use the identity,

$$\int_x^\infty \exp\left(-\frac{t^2}{2}\right) dt \leq \frac{1}{x} \exp\left(-\frac{x^2}{2}\right) \quad (17)$$